



The Future of Linux and Emerging Trends

What's next?

For decades, Linux has remained at the forefront of technological revolutions, evolving from a hobbyist operating system to a cornerstone of enterprise computing, cloud infrastructure, and embedded systems. As the world leans further into digital transformation, the role of Linux continues to expand, offering scalability, security, and open-source flexibility. The future of Linux promises to be deeply intertwined with emerging technologies such as artificial intelligence, edge computing, and quantum computing. In this chapter, we explore the trajectory of Linux, its adaptation to new hardware, security advancements, enterprise adoption, governance challenges, and predictions for its future evolution. Linux has undergone numerous transformations since its inception in 1991 by Linus

Torvalds. Originally developed as a Unix-like operating system, it quickly garnered attention from developers and enthusiasts due to its open-source nature. Over the years, Linux's reliability, security, and community-driven development model have enabled its adoption in everything from personal computers to critical infrastructure. Understanding these evolutionary trends provides insight into the potential future pathways for Linux.

The Evolution of Linux

The Growing Role of Linux in Cloud and Edge Computing

Linux is the backbone of cloud computing, with platforms